CURRICULUM VITAE

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B.Tech - Computer Science and Engineering
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Educational Qualifications

Graduation (NIT Durgapur)

Semester	1	2	3	4
SGPA	7.26	8.65	8.08	7.73
CGPA	7.26	7.91	7.97	7.91

Higher Secondary and Secondary

- **Class XII (CBSE)** - Delhi Public School Durgapur – **87%**
- **Class X (CBSE)** – Delhi Public School Durgapur – **93%**

Academic Achievements

- Scored **98.1 percentile** in **JEE Mains**, secured rank 22K among 16 lakh+ students.
- Scored **10 grades** in **Mathematics, DSA, OOPs** and **9** in **Theory of Computation**.
- **Certificate of Merit** from CBSE for scoring 100/100 in SST; multiple honor cards in school.
- School rank 3 and International Rank 2024 in IMO (SOF) in class 10.
- Certified in **Data Science**, **Gen AI**, **Decode C++** (PW Skills, Udemy).

Skills Summary

Languages: Python, C++, SQL
Frameworks/Libraries: TensorFlow, Flask, Streamlit, OpenCV
AI/ML: Deep Learning, NLP, Transformers, GANs, YOLO, Whisper, Computer Vision
Tools: Git, VS Code, Anaconda, Langchain
Coding: Solved over 600 problems on **LeetCode** with a rating of **1773 (Top 8%)**.

Internships

Project Intern – Techsaksham (Microsoft + SAP)
Remote — Jan 2025 – Feb 2025
Developed a multi-disease prediction system using **Python** and **Scikit-learn**. **Deployed** with **Flask and Streamlit**; achieved **88% accuracy** using cross-validation.

Project Intern – 1Stop.ai
Remote — Apr 2025 – Jul 2025
Built a hate speech detector using **NLTK**, **Scikit-learn**, **DecisionTreeClassifier** with **EDA using**

Seaborn, Numpy. Also developed a **Heart Disease mortality predictor** using **Scikit-learn**, evaluated using **confusion matrix**
metrics.

Selected Projects

Advanced Event Management System
Tech Stack: Python, SQLite, Streamlit
Fully **deployed on Streamlit Cloud** — designed a modular DBMS project with **full CRUD operations**, **search/custom queries**, and **role-specific views** for Attendees, Hosts, Events, Bills, and Suppliers. Supports **automated billing**, **data integrity** using **triggers/views**, and **real-time responsiveness**.
GitHub **Live Demo**

Voice Assistant
Tech Stack: Flask, Langchain, Python, SpeechRecognition, VLC, OpenAI/Groq API
Built a **voice assistant** to **answer queries, play music, set reminders**, and fetch weather. Integrated **Langchain+Groq** for **NLP intent detection** and **voice I/O**. Added **YouTube playback** with **yt-dlp** and **python-vlc**. Provided a **demo video** showcasing features.
GitHub **Demo Video**

English-French Translator
Tech Stack: Python, TensorFlow/Keras, Transformer
Developed a **Transformer encoder-decoder model** with **multi-head attention** and **positional encoding**. Achieved **high BLEU scores** and **faster, more accurate translation** than RNNs.
GitHub

Grammatical Error Correction
Tech Stack: Python, TensorFlow/Keras, Hugging Face T5, NLP
Fine-tuned **T5 Seq2Seq** (222M+ params) for **grammar correction**, with **BLEU evaluation**, **custom loss**, and **learning rate schedules**. Also tested **zero-shot inference**.
GitHub

Sign Language Detector
Tech Stack: Python, OpenCV, TensorFlow/Keras, MediaPipe
Real-time **hand gesture recognition** using **MediaPipe** + **LSTM sequence classification**. **Custom dataset**, **live predictions**, and **confidence filtering**.
GitHub

Malaria Detection
Tech Stack: Python, TensorFlow/Keras, OpenCV, Albumentations
Built CNNs (**LeNet, Sequential API**) to classify **malaria-infected cells**. Used **advanced augmentations** (**MixUp, CutMix**) and **metrics visualization** (**ROC, confusion matrix, W&B**). Achieved **87% val acc**, **95% AUC**.
GitHub